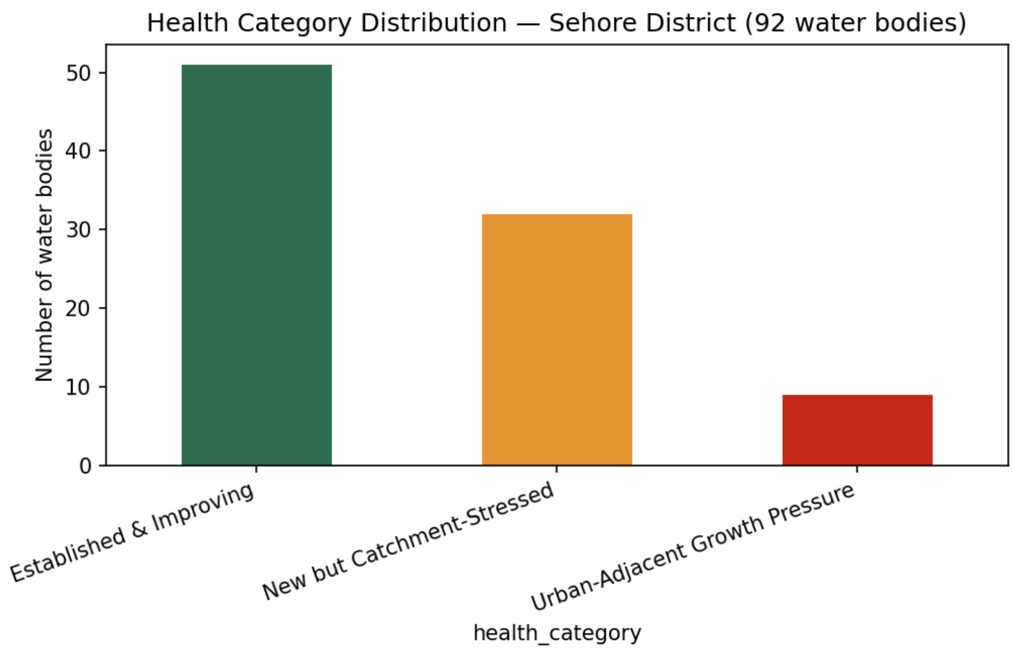
Water Body Health Card

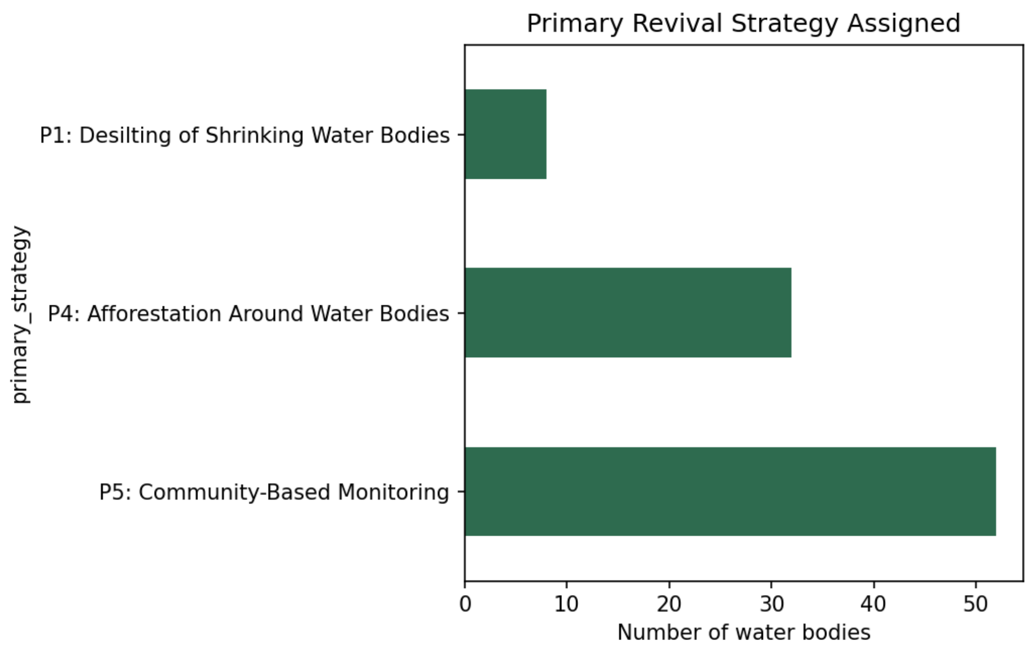
Sehore District, Madhya Pradesh

Automated assessment generated from Sentinel-2 NDWI/NDVI time series, MODIS/Dynamic World land cover, and k-means health clustering (DSN3099 EPICS Project).

Executive Summary

A total of 92 distinct water bodies were identified and assessed across Sehore district using multi-temporal satellite imagery (2017-2024). Of these, 51 (55%) are classified as 'Established & Improving', 31 (34%) are new since 2017, and 8 water bodies show a net decline in surface water area and are flagged as urgent priority for desilting intervention.





Priority Tier 1 — Urgent (Net Water Loss)

These water bodies show a net decline in surface water area since 2017, despite in some cases sitting within an otherwise stable cluster average. These are the strongest candidates for field verification and immediate desilting intervention.

Water Body 000013

Water Body 000013 covers approximately 0.57 hectares and falls into the 'Established & Improving' category. Surface water area has declined by 0.4% since 2017. Surrounding catchment vegetation has improved (NDVI change: 0.011), within a catchment that is 42% cropland and 0% urban/built. This water body requires urgent attention. Recommended primary strategy: P1: Desilting of Shrinking Water Bodies.

Water Body 000033

Water Body 000033 covers approximately 0.81 hectares and falls into the 'Established & Improving' category. Surface water area has declined by 0.3% since 2017. Surrounding catchment vegetation has improved (NDVI change: 0.016), within a catchment that is 0% cropland and 0% urban/built. This water body requires urgent attention. Recommended primary strategy: P1: Desilting of Shrinking Water Bodies.

Water Body 000034

Water Body 000034 covers approximately 5.79 hectares and falls into the 'Established & Improving' category. Surface water area has declined by 0.2% since 2017. Surrounding catchment vegetation has improved (NDVI change: 0.024), within a catchment that is 0% cropland and 0% urban/built. This water body requires urgent attention. Recommended primary strategy: P1: Desilting of Shrinking Water Bodies.

Water Body 000036

Water Body 000036 covers approximately 3.22 hectares and falls into the 'Established & Improving' category. Surface water area has declined by 0.1% since 2017. Surrounding catchment vegetation has improved (NDVI change: 0.011), within a catchment that is 4% cropland and 0% urban/built. This water body requires urgent attention. Recommended primary strategy: P1: Desilting of Shrinking Water Bodies.

Water Body 000043

Water Body 000043 covers approximately 0.64 hectares and falls into the 'Established & Improving' category. Surface water area has declined by 0.1% since 2017. Surrounding catchment vegetation has improved (NDVI change: 0.014), within a catchment that is 28% cropland and 0% urban/built. This water body requires urgent attention. Recommended primary strategy: P1: Desilting of Shrinking Water Bodies.

Water Body 000005

Water Body 000005 covers approximately 0.61 hectares and falls into the 'Established & Improving' category. Surface water area has declined by 0.1% since 2017. Surrounding catchment vegetation has declined (NDVI change: -0.036), within a catchment that is 73% cropland and 3% urban/built. This water body requires urgent attention. Recommended primary strategy: P1: Desilting of Shrinking Water Bodies.

Water Body 000044

Water Body 000044 covers approximately 4.35 hectares and falls into the 'Established & Improving' category. Surface water area has declined by 0.1% since 2017. Surrounding catchment vegetation has improved (NDVI change: 0.049), within a catchment that is 92% cropland and 3% urban/built. This water body requires urgent attention. Recommended primary strategy: P1: Desilting of Shrinking Water Bodies.

Water Body 000014

Water Body 000014 covers approximately 28.63 hectares and falls into the 'Established & Improving' category. Surface water area has declined by 0.0% since 2017. Surrounding catchment vegetation has declined (NDVI change: -0.002), within a catchment that is 88% cropland and 6% urban/built. This water body requires urgent attention. Recommended primary strategy: P1: Desilting of Shrinking Water Bodies.

Priority Tier 2 — Multiple Compounding Pressures

ID	Area (ha)	Category	Primary Strategy	# Strategies
000010	9.0	Established & Improving	P5: Community-Based Monitoring	3
000018	81.0	Established & Improving	P5: Community-Based Monitoring	3
000017	97.8	Established & Improving	P5: Community-Based Monitoring	4
000016	8.4	Established & Improving	P5: Community-Based Monitoring	3
00000d	2.3	Established & Improving	P5: Community-Based Monitoring	3
00000e	14.7	Established & Improving	P5: Community-Based Monitoring	3
000006	8.8	Urban-Adjacent Growth Pressure	P5: Community-Based Monitoring	5
000003	28.9	Established & Improving	P5: Community-Based Monitoring	3
000023	45.2	Established & Improving	P5: Community-Based Monitoring	4
000015	5.5	Urban-Adjacent Growth Pressure	P5: Community-Based Monitoring	4
000008	1.2	Urban-Adjacent Growth Pressure	P5: Community-Based Monitoring	4
00003a	1.2	New but Catchment-Stressed	P4: Afforestation Around Water Bodies	3
000040	0.6	Urban-Adjacent Growth Pressure	P5: Community-Based Monitoring	3
000041	0.9	Urban-Adjacent Growth Pressure	P5: Community-Based Monitoring	3
000004	27.0	Established & Improving	P5: Community-Based Monitoring	4
000002	5.6	Established & Improving	P5: Community-Based Monitoring	4
000059	261.5	Established & Improving	P5: Community-Based Monitoring	3
00004f	1.8	Urban-Adjacent Growth Pressure	P5: Community-Based Monitoring	4

Priority Tier 3 — Standard Monitoring

ID	Area (ha)	Category	Primary Strategy
000011	0.6	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000019	4.7	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
00000f	0.9	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
00000c	1.4	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
00000a	3.9	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000012	3.9	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
00001f	1.1	Urban-Adjacent Growth Pressure	P5: Community-Based Monitoring
00001e	0.6	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
00001d	4.1	Established & Improving	P5: Community-Based Monitoring
00001c	7.7	Established & Improving	P5: Community-Based Monitoring
00001b	10.5	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000020	6.7	Established & Improving	P5: Community-Based Monitoring
000021	1.2	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000022	0.7	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000028	1.0	New but Catchment-Stressed	P4: Afforestation Around Water Bodies

ID	Area (ha)	Category	Primary Strategy
000029	1.1	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
00002a	0.7	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
00002b	0.8	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000024	0.7	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000025	2.5	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000026	355.5	Established & Improving	P5: Community-Based Monitoring
000027	46.3	Established & Improving	P5: Community-Based Monitoring
00002f	16.4	Established & Improving	P5: Community-Based Monitoring
00002e	1.5	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
00002d	852.5	Established & Improving	P5: Community-Based Monitoring
00002c	0.6	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000032	17.9	Established & Improving	P5: Community-Based Monitoring
000031	5.1	Established & Improving	P5: Community-Based Monitoring
000035	191.2	Established & Improving	P5: Community-Based Monitoring
000030	0.8	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000037	101.7	Established & Improving	P5: Community-Based Monitoring
000038	2.5	Established & Improving	P5: Community-Based Monitoring
00001a	1.7	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
00000b	0.7	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000009	1.2	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000001	23.6	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000000	70.1	Established & Improving	P5: Community-Based Monitoring
000007	5.1	Established & Improving	P5: Community-Based Monitoring
00003f	104.6	Established & Improving	P5: Community-Based Monitoring
00003e	30.0	Established & Improving	P5: Community-Based Monitoring
00003d	13.5	Established & Improving	P5: Community-Based Monitoring
00003c	1.4	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
00003b	1.2	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000039	1.9	Urban-Adjacent Growth Pressure	P5: Community-Based Monitoring
000046	92.4	Established & Improving	P5: Community-Based Monitoring
000042	2.2	Established & Improving	P5: Community-Based Monitoring
000047	3.4	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000048	3.9	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
00004a	39.8	Established & Improving	P5: Community-Based Monitoring
000049	33.7	Established & Improving	P5: Community-Based Monitoring
00004b	1.7	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
00004c	17.2	Established & Improving	P5: Community-Based Monitoring
00004d	94.4	Established & Improving	P5: Community-Based Monitoring

ID	Area (ha)	Category	Primary Strategy
000045	9.2	Established & Improving	P5: Community-Based Monitoring
00004e	5.4	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000050	7.5	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000052	6.4	Established & Improving	P5: Community-Based Monitoring
000051	10.5	Urban-Adjacent Growth Pressure	P5: Community-Based Monitoring
000054	0.6	Established & Improving	P5: Community-Based Monitoring
000055	0.5	New but Catchment-Stressed	P4: Afforestation Around Water Bodies
000056	125.7	Established & Improving	P5: Community-Based Monitoring
000053	35.0	Established & Improving	P5: Community-Based Monitoring
000057	59.1	Established & Improving	P5: Community-Based Monitoring
000058	246.5	Established & Improving	P5: Community-Based Monitoring
00005a	60.4	Established & Improving	P5: Community-Based Monitoring
00005b	56.6	Established & Improving	P5: Community-Based Monitoring

Methodology & Limitations

Water body boundaries were derived from Sentinel-2 NDWI thresholding (2024) and validated against 18 manually labeled ground-truth points near VIT Bhopal campus (77.8% accuracy, kappa 0.48 for both the NDWI threshold and a Random Forest classifier — no measurable difference at this sample size and location; results should not be assumed to generalize district-wide). Catchment buffers were scaled per water body (5x each body's own equivalent radius, clamped 100m-2000m) rather than using a fixed buffer for all sites. Health categories were derived via k-means clustering (k=3, chosen by silhouette score, not assumed in advance) on standardized features including log-transformed area, capped water-area percent change, catchment NDVI trend, and land-use composition. Strategy assignment combines cluster-level defaults with individual per-water-body override rules. 31 of 92 water bodies (34%) show near-zero detected water in 2017 and are treated as new construction rather than assigned an unstable percent-change value.